BULGARIAN NATIONAL OLYMPIAD IN MATHEMATICS

Third round

1995

Problem 1. Let p and q be positive numbers such that the parabola $y = x^2 - 2px + q$ has no common point with the x-axis. Prove that there exist points A and B on the parabola such that the segment AB is parallel to the x-axis and $\angle AOB = 90^\circ$ (O is the coordinate origin) if and only if $p^2 < q \leq \frac{1}{4}$. Find the values of p and q for which the points A and B are defined in an unique way.

Problem 2. Let $A_1A_2A_3A_4A_5A_6A_7$, $B_1B_2B_3B_4B_5B_6B_7$, $C_1C_2C_3C_4C_5C_6C_7$ be regular heptagons with areas S_A, S_B and S_C, respectively. Let $A_1A_2 = B_1B_3 = C_1C_4$. Prove that

$$\frac{1}{2} < \frac{S_B + S_C}{S_A} < 2 - \sqrt{2}.$$

Problem 3. Let $n > 1$ be an integer. Find the number of the permutations $(a_1, a_2, \ldots, a_n)$ of the numbers $1, 2, \ldots, n$ with the following property: there exists only one index $i \in \{1, 2, \ldots, n-1\}$ such that $a_i > a_{i+1}$.

Problem 4. Let $n \geq 2$ and $0 \leq x_i \leq 1$ for $i = 1, 2, \ldots, n$. Prove the inequality

$$(x_1 + x_2 + \cdots + x_n) - (x_1x_2 + x_2x_3 + \cdots + x_{n-1}x_n + x_nx_1) \leq \left[\frac{n}{2}\right].$$

When is there an equality?

Problem 5. The points A_1, B_1, C_1 lie on the sides BC, CA, AB of the triangle ABC respectively and the lines AA_1, BB_1, CC_1 have a common point M. Prove that if the point M is center of gravity of $\triangle A_1B_1C_1$, then M is the center of gravity of $\triangle ABC$.

Problem 6. Find all pairs of positive integers (x, y) for which $\frac{x^2 + y^2}{x - y}$ is an integer which is a divisor of 1995.

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